

# Conversational AI Customer Intent Classification

Annotation Guidelines — ML Data Annotation Operations Project

## 1. Purpose

This document defines the rules for consistently assigning one primary intent label to each synthetic customer-service conversation. The goal is to produce reliable, auditable labels suitable for machine-learning training and evaluation.

**Annotation principle:** label what the customer is asking for or reporting, not what the annotator assumes the company should do.

## 2. Dataset Scope

| Item             | Specification                            |
|------------------|------------------------------------------|
| Data type        | Synthetic customer-service conversations |
| Expected records | 5,000                                    |
| Intent classes   | 20                                       |
| Labeling task    | Single-label classification              |
| Language         | English                                  |
| Primary output   | One intent per conversation              |

## 3. General Annotation Rules

- Read the entire customer message** before assigning a label.
- Identify the customer's primary intent**, meaning the main request or problem the customer wants resolved.
- Do not infer facts** that are not stated or reasonably established by the message.
- Use the most specific applicable intent.** Do not choose a broad category when a specific category clearly applies.
- Use the intent definitions consistently.** Personal intuition should not override the documented rules.
- Ignore emotional tone** unless it changes the actual request. Anger about a late delivery is still `delivery_delay`.
- Do not classify based on individual keywords alone.** Consider the meaning and context of the complete message.
- One primary label only.** If multiple issues are present, apply the multi-intent rule below.
- Escalate genuine ambiguity** when the message cannot be resolved using these rules.

## 4. Multi-Intent Rule

This project uses single-label classification. When a message contains multiple issues, select the **primary customer request**. Prefer the issue that is most directly stated as the requested action or the main reason for contacting support.

**Example:** “My parcel is late and I want to cancel the order.”

If the customer is primarily requesting cancellation, use `order_cancellation`. If the customer is primarily reporting the missed delivery and asking for an explanation, use `delivery_delay`.

If the primary intent cannot be determined reliably, mark the record for review rather than guessing.

## 5. Ambiguity and Escalation

- Use `<b>NEEDS_REVIEW</b>` during internal QA when two labels remain equally plausible after applying the guidelines.
- Record the reason for the disagreement so the guideline owner can decide whether a rule needs refinement.
- Do not create a new intent during annotation. New categories must be approved through taxonomy governance.
- After a taxonomy clarification is approved, update the guideline and re-review affected examples.

## 6. Intent Definitions and Decision Rules

### INT-001 — order\_status

<b>Definition:</b> Current status/progress of an existing order.

<b>Use when:</b> General status update; not specifically tracking, ETA, or delay.

<b>Do not use when:</b> Tracking, delivery estimate, or late delivery.

<b>Example:</b> “Can you check my order status?”

### INT-002 — delivery\_delay

<b>Definition:</b> Expected delivery is late or the promised window has passed.

<b>Use when:</b> Customer explicitly reports a late/missed delivery.

<b>Do not use when:</b> Do not use if only asking for location or ETA before deadline.

<b>Example:</b> “My package was due yesterday and still hasn’t arrived.”

<b>Decision cue:</b> Missed/passed delivery deadline → delivery\_delay; location or tracking request without an established missed deadline → delivery\_tracking; future expected arrival/ETA question → delivery\_estimate.

### INT-003 — delivery\_estimate

<b>Definition:</b> Customer asks when an order is expected to arrive.

<b>Use when:</b> ETA, expected date, or delivery duration.

<b>Do not use when:</b> Do not use for an already-late delivery.

<b>Example:</b> “When should my package arrive?”

<b>Decision cue:</b> Missed/passed delivery deadline → delivery\_delay; location or tracking request without an established missed deadline → delivery\_tracking; future expected arrival/ETA question → delivery\_estimate.

### INT-004 — delivery\_tracking

<b>Definition:</b> Customer wants shipment location, tracking, scans, or transit status.

<b>Use when:</b> Tracking/location request without an established missed deadline.

<b>Do not use when:</b> Do not use when the customer says the delivery is already late.

<b>Example:</b> “Where is my parcel right now?”

<b>Decision cue:</b> Missed/passed delivery deadline → delivery\_delay; location or tracking request without an established missed deadline → delivery\_tracking; future expected arrival/ETA question → delivery\_estimate.

### INT-005 — return\_request

<b>Definition:</b> Customer wants to return a received product or asks how to start a return.

<b>Use when:</b> Starting or understanding a product return.

<b>Do not use when:</b> Do not use for pre-shipment cancellation or refund status.

<b>Example:</b> “How do I return this item?”

<b>Decision cue:</b> Stop an order before completion/shipment → order\_cancellation; send an already-received product back → return\_request; ask about money already expected from a refund → refund\_status.

### **INT-006 — refund\_status**

<b>Definition:</b> Customer asks about money expected from an existing refund/return.

<b>Use when:</b> Refund processing status or expected refund timing.

<b>Do not use when:</b> Do not use for starting a return.

<b>Example:</b> “Has my refund been processed?”

<b>Decision cue:</b> Stop an order before completion/shipment → order\_cancellation; send an already-received product back → return\_request; ask about money already expected from a refund → refund\_status.

### **INT-007 — payment\_failure**

<b>Definition:</b> Payment attempt is declined, rejected, or fails during checkout.

<b>Use when:</b> Unsuccessful payment attempt.

<b>Do not use when:</b> Do not use for duplicate or already-successful charges.

<b>Example:</b> “My card payment keeps failing.”

### **INT-008 — duplicate\_charge**

<b>Definition:</b> Same purchase appears to have been charged more than once.

<b>Use when:</b> Two or more charges for one purchase.

<b>Do not use when:</b> Do not use for a single unexpected charge.

<b>Example:</b> “I was charged twice for one order.”

### **INT-009 — order\_cancellation**

<b>Definition:</b> Customer wants to stop an existing order before fulfillment/shipment.

<b>Use when:</b> Cancelling a one-time order.

<b>Do not use when:</b> Do not use for returning a received item or cancelling a subscription.

<b>Example:</b> “Can you cancel my order?”

<b>Decision cue:</b> Stop an order before completion/shipment → order\_cancellation; send an already-received product back → return\_request; ask about money already expected from a refund → refund\_status.

### **INT-010 — damaged\_item**

<b>Definition:</b> Received product is broken, cracked, damaged, or defective on arrival.

<b>Use when:</b> Physical damage is the primary issue.

<b>Do not use when:</b> Do not use for wrong or missing items.

<b>Example:</b> “The product arrived broken.”

<b>Decision cue:</b> Correct item but physically damaged → damaged\_item; different item received → wrong\_item; expected item absent → missing\_item.

### **INT-011 — wrong\_item**

<b>Definition:</b> Received product does not match what was ordered.

<b>Use when:</b> Wrong model, product, size, color, or variant.

<b>Do not use when:</b> Do not use when the correct product is damaged or missing.

<b>Example:</b> “I received the wrong model.”

<b>Decision cue:</b> Correct item but physically damaged → damaged\_item; different item received → wrong\_item; expected item absent → missing\_item.

### **INT-012 — missing\_item**

<b>Definition:</b> One or more ordered products are absent from a delivered package.

<b>Use when:</b> Package arrived but an ordered item is missing.

<b>Do not use when:</b> Do not use when a different item was delivered.

<b>Example:</b> “One item from my order is missing.”

<b>Decision cue:</b> Correct item but physically damaged → damaged\_item; different item received → wrong\_item; expected item absent → missing\_item.

### **INT-013 — account\_access**

<b>Definition:</b> Customer cannot sign in or access their account.

<b>Use when:</b> General login/access problem.

<b>Do not use when:</b> Do not use for explicit password recovery or suspected compromise.

<b>Example:</b> “I can't sign in to my account.”

<b>Decision cue:</b> General inability to sign in → account\_access; forgotten/reset password → password\_reset; suspected unauthorized access or suspicious activity → account\_security.

### **INT-014 — password\_reset**

<b>Definition:</b> Customer forgot a password or explicitly requests password recovery/reset.

<b>Use when:</b> Password reset or forgotten-password request.

<b>Do not use when:</b> Do not use for generic login problems or security incidents.

<b>Example:</b> “I forgot my password. How can I reset it?”

<b>Decision cue:</b> General inability to sign in → account\_access; forgotten/reset password → password\_reset; suspected unauthorized access or suspicious activity → account\_security.

### **INT-015 — account\_security**

<b>Definition:</b> Customer reports suspicious login, unauthorized change, or possible account compromise.

<b>Use when:</b> Security incident or suspected unauthorized access.

<b>Do not use when:</b> Do not use for ordinary login/password problems.

<b>Example:</b> “I don't recognize this login.”

<b>Decision cue:</b> General inability to sign in → account\_access; forgotten/reset password → password\_reset; suspected unauthorized access or suspicious activity → account\_security.

### **INT-016 — address\_change**

<b>Definition:</b> Customer wants to change/correct the shipping address for an order.

<b>Use when:</b> Delivery-address modification.

<b>Do not use when:</b> Do not use for unrelated profile changes.

<b>Example:</b> “I entered the wrong shipping address.”

### **INT-017 — subscription\_cancel**

**Definition:** Customer wants to stop a recurring subscription or prevent renewal.

**Use when:** Recurring membership cancellation.

**Do not use when:** Do not use for subscription billing questions without cancellation intent.

**Example:** “Please cancel my subscription.”

### **INT-018 — subscription\_billing**

**Definition:** Customer asks about a recurring subscription charge or billing amount.

**Use when:** Subscription billing explanation/dispute.

**Do not use when:** Do not use for subscription cancellation.

**Example:** “Why was I charged for my subscription?”

### **INT-019 — promo\_code**

**Definition:** Customer asks about a promotional code or reports a coupon/discount code problem.

**Use when:** Promo-code usage, rejection, or discount not applying.

**Do not use when:** Do not use for general product pricing questions.

**Example:** “My discount code isn’t working.”

### **INT-020 — product\_information**

**Definition:** Customer requests product features, specifications, contents, options, or compatibility information.

**Use when:** Informational product question.

**Do not use when:** Do not use for problems with a received item.

**Example:** “What features does this product have?”

## 7. High-Confusion Intent Pairs

| Confusing pair                              | How to distinguish                                                                                                            |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| delivery_tracking vs delivery_delay         | Tracking asks where the shipment is. Delay states that the expected delivery date/window has been missed.                     |
| delivery_estimate vs delivery_delay         | Estimate asks when it should arrive. Delay indicates that the expected arrival has already passed.                            |
| return_request vs order_cancellation        | Return applies to sending back a received product. Cancellation applies to stopping an existing order before it ships.        |
| return_request vs refund_status             | Return is about initiating/understanding the return. Refund status is about money expected after the refund/return.           |
| order_status vs delivery_tracking           | Order status is a broad order-progress request. Tracking is specifically about shipment location, scans, or tracking numbers. |
| account_access vs password_reset            | Account access is a general login problem. Password reset is explicitly about a forgotten/reset password.                     |
| account_access vs account_security          | Login difficulty alone is account_access. Suspicious activity or suspected unauthorized access is account_security.           |
| damaged_item vs wrong_item                  | Damaged item means the expected product arrived damaged. Wrong item means a different product/variant was delivered.          |
| wrong_item vs missing_item                  | Wrong item means something different was delivered. Missing item means an expected item was absent.                           |
| payment_failure vs duplicate_charge         | Payment failure is an unsuccessful transaction. Duplicate charge is multiple successful-looking charges for the same item.    |
| subscription_cancel vs subscription_billing | Cancellation requests stopping the subscription. Billing asks about a recurring charge or amount.                             |

## 8. Annotation Quality Standards

- Annotators must follow the written taxonomy rather than relying on keyword matching.
- The same scenario should receive the same label regardless of writing style, punctuation, or emotional tone.
- Short messages must still be labeled when the intent is sufficiently clear.
- Messages with insufficient evidence should be escalated rather than assigned arbitrarily.
- Quality reviewers should sample annotations and compare them against an approved gold-standard key.
- Repeated disagreements involving the same intent pair should trigger guideline review.

## 9. Compliance and Privacy Annotation

Although the project dataset is synthetic, the operational workflow includes a compliance check. Annotators and reviewers should flag apparent personal or sensitive information if it occurs, including phone numbers, email addresses, physical addresses, payment-card details, passwords, authentication codes, or other information that should not be unnecessarily exposed in a training dataset.

**Important:** Compliance status is separate from intent. A message can have both an intent label and a compliance flag during the audit stage.

## 10. Recommended Annotation Output

| Field            | Purpose                                           |
|------------------|---------------------------------------------------|
| conversation_id  | Unique conversation identifier                    |
| customer_message | Original customer text                            |
| annotated_intent | One approved intent label                         |
| annotator_id     | Identifier for the annotator/reviewer             |
| review_status    | approved / needs_review / corrected               |
| review_reason    | Reason when a label requires review or correction |

## 11. Version Control

Guideline version: 1.0<br/>Task: Customer-service intent classification<br/>Taxonomy size: 20 intents<br/>Annotation policy: single primary intent<br/>Status: Initial production draft for annotation and QA